

Mayank Agarwal

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SUMMARY

Software Engineer with experience building scalable backend systems using **Java, Spring Boot, AWS, PostgreSQL, and REST APIs**. Experienced in designing high-performance financial data services, optimizing database queries, implementing caching mechanisms, and handling large-scale datasets. Passionate about distributed systems, backend engineering, and machine learning.

EXPERIENCE

S&P Global

June 2025 – Present

Software Development - Apprentice

Remote/Office

- Designed and implemented a backend caching mechanism for ETF and Index Basket data retrieval, reducing API response latency by **5%** and improving overall client-side performance.
- Developed and maintained Spring Boot REST APIs supporting large-scale financial data workflows involving millions of index and ETF records.
- Designed optimized pagination and sorting strategies for large datasets, significantly improving query performance and reducing database response time.
- Deployed and monitored backend applications using **AWS ECS, Elastic Beanstalk, and CloudWatch**, ensuring reliable production deployments and system observability.
- Optimized SQL queries, indexing strategies, and pagination mechanisms to improve query execution performance for large-scale financial datasets.
- Collaborated with cross-functional teams to design and implement backend features while following clean architecture and industry best practices.

PROJECTS

Medicinal Leaf Classification | *Python, Machine Learning, Computer Vision* | [Link](#)

Aug. 2024 – May 2025

- Built a **ML-based medicinal leaf classification** system using **Canny Edge Detection, Inception Learning, and Ensemble Learning** models.
- Achieved **97% model accuracy** on a dataset of **30,000+ plant leaf images**.
- Optimized **model inference** and application performance to improve user experience and reduce processing time.
- Designed the project following a modular architecture to support future model deployment and scalability
- Optimized inference pipeline to reduce prediction time and improve application responsiveness

Game Recommendation Engine | *Node.js, Express.js, REST APIs* | [Link](#)

Aug. 2023 – Oct. 2023

- Built a **full-stack web application** for gamers and developers to fetch game performance information.
- Developed **backend APIs** and integrated **search functionality** with **live stream support**.
- Enhanced user experience by **30%** and improved overall application optimization by **10%**.

TECHNICAL SKILLS

Languages: Java, Kotlin, Python, C, C++

Backend & Frameworks: Spring Boot, REST APIs, NodeJS, Express JS, Hibernate, JPA, JDBC, JWT, Maven, OAuth 2.0

AI & LLM Technologies: Claude API, Prompt engineering, AI-Assisted Development

Databases: SQL, NoSQL, PostgreSQL, SQL Server Management (SSM)

Cloud & DevOps: AWS, Amazon ECS, Elastic Beanstalk, CloudWatch, Git, CI/CD

Core Concepts: OOPs, Data Structures & Algorithms, Streams, Multithreading, NIO, Swing, REST Architecture

ACHIEVEMENTS

- Solved **600+ coding problems** across **Leetcode, Codeforces, and CSES** platforms.
- Ranked among the **Top 10 teams** at **Hack X Treme Hackathon** organized at **IIT Delhi** by Dev Club.

EDUCATION

Lovely Professional University

B.Tech in Computer Science and Engineering (CGPA: 8.20)

Punjab, India

2021 – 2025